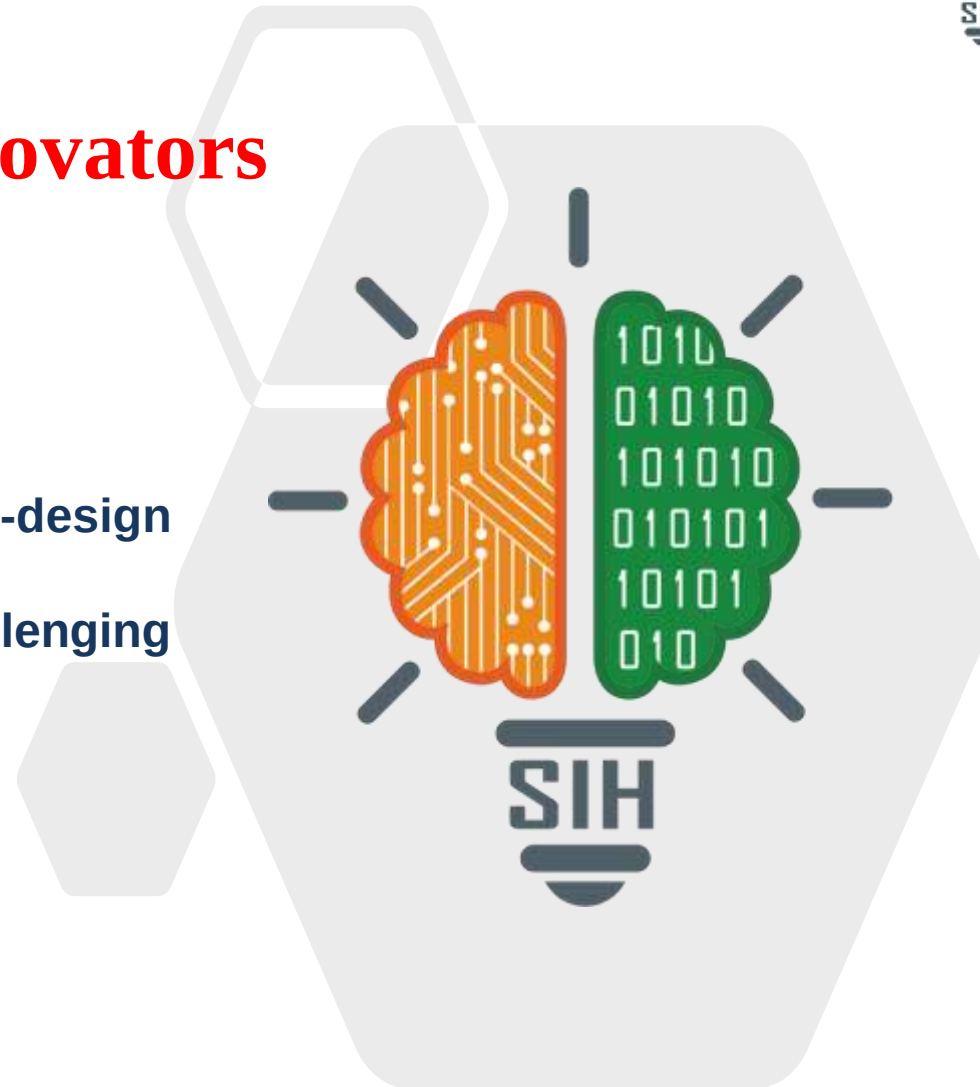


SMART INDIA HACKATHON 2026

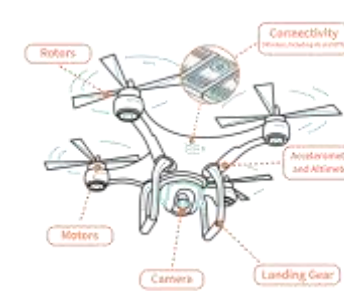


The Innovators

- Problem Statement ID –SIH26218
- Problem Statement Title-Student Innovation:-design a drone and robots that help in challenging situation
- Theme-Robotics and Drones
- PS Category-Hardware
- Team ID-SIH022
- Team Name-The Innovators



IDEA TITLE



PROBLEM



The main problem due to disasters:-

- **Health risk** :- Dirty water pollutes drinking supplies which causes problem
- **Food shortage** :- Deep water ruins farm fields, spoils stored food and kills livestock
- **Rescue on time**:- Due to not finding the people it causes death.

#FLYER-SAVIOUR#

who saves life

OUR SOLUTION

We make a drone for these type of situation by which

1. we can easily find where the people is hiding and rescue them
2. We can also supply medical support to those people
3. We can also supply food, water and other essential things for their survival.

#WHY DIFFERENT#

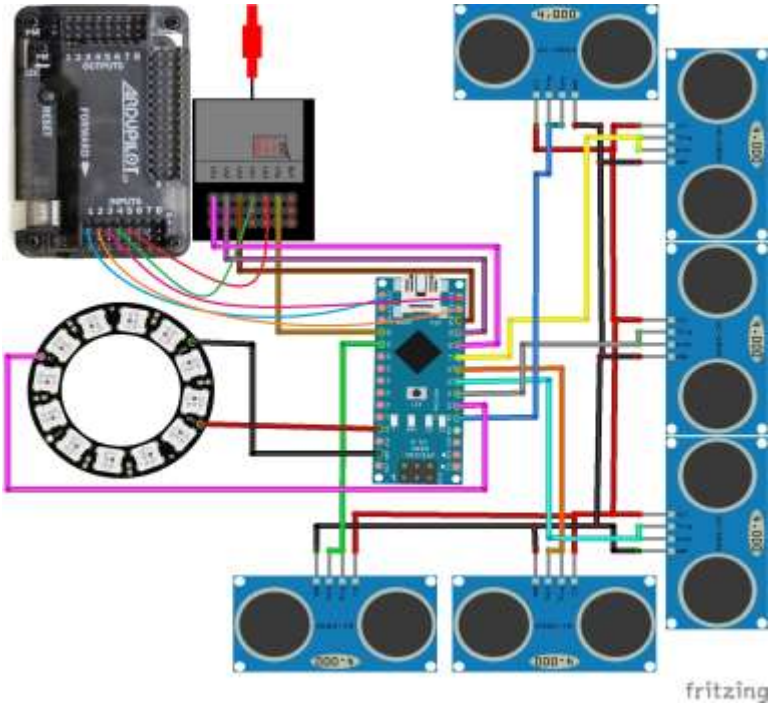
- We use a ultra sonic sensor which help to remove all obstacles it self.
- We make it very stable which help to supply all type of essential thing to people effectively.
- We also protect the camera from damage by covering it.

KEY VALUE PROPOSTION

- Work for all the rescue operations and for helping government.
- Accurate working, fast working.
- Economically feasible for all the authorities.

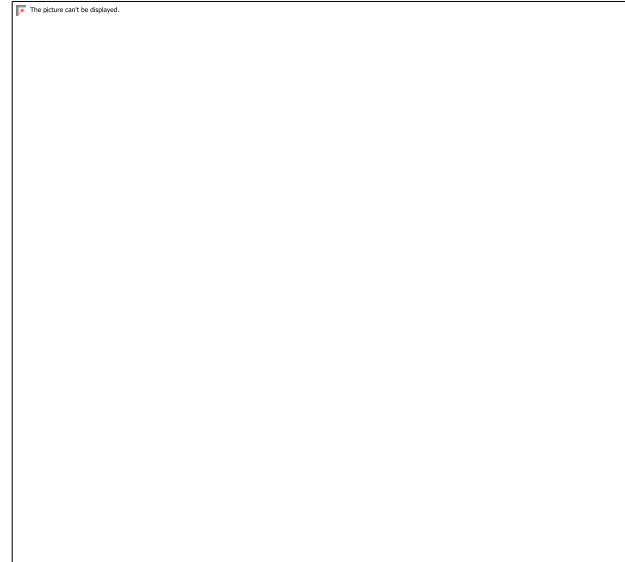
TECHNICAL APPROACH

The Innovators

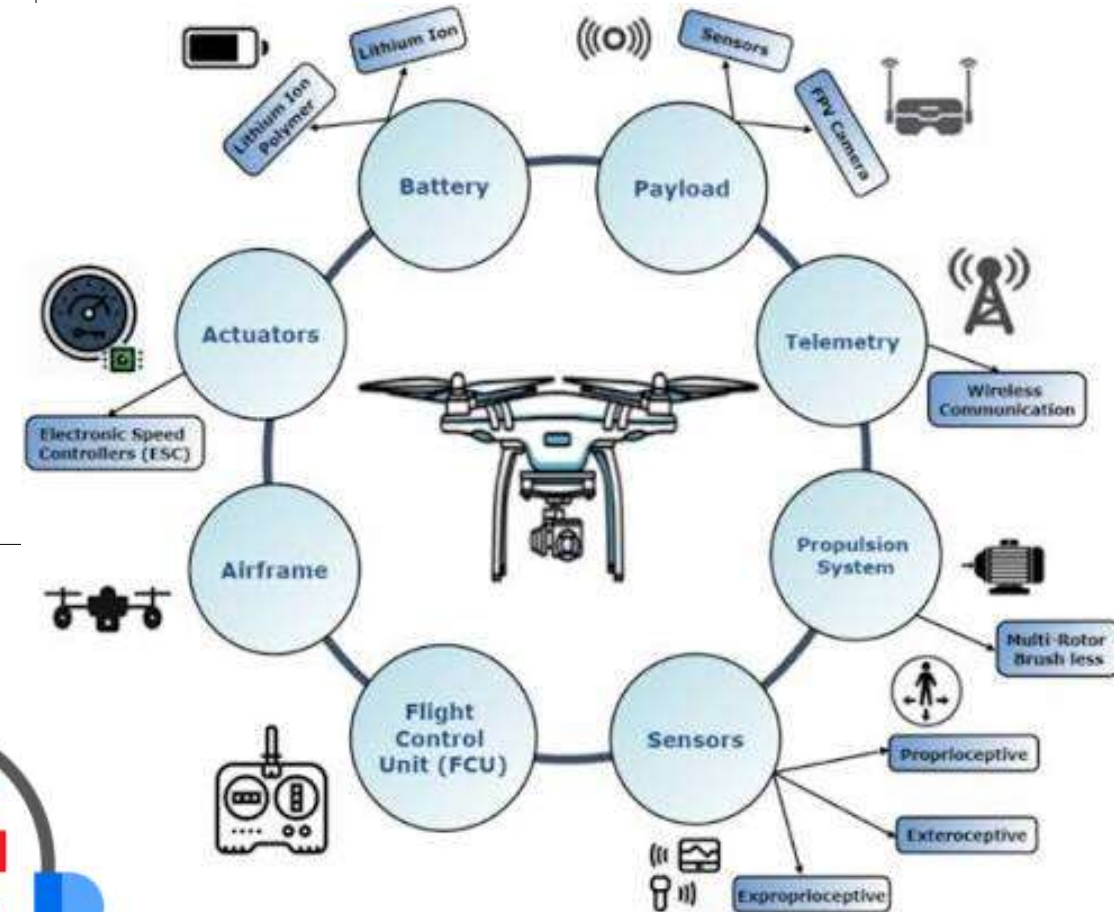


Connections of sensor

Tech Stack Used:- Arduino IDE, microcontroller & processor, sensor, power supply, Robot operating system(ROS).



Balancing process



Process Flowchart

Risk assessment and mitigation

Problem:-Sudden Obstacles



Solution:-Uses of different sensors

Problem:-environmental damage



Solution:-pre flight checking and use of hard and light weight material

Problem:-Lost of signal



Solution:-Uses of effective transmitter and receiver



Resource Requirement

- Its required core electronic component like:-
- Frame
- Motor & propellers
- Electronic Speed Controller(ESCs)
- Flight controller
- Battery & power distribution
- Radio system & Optional Add-ons

Economical Approach

- Brushed(coreless) motor over Brushless
- Use of Arduino Nano or an ESP32 board combined with an MPU-6050 sensor
- Modular kits
- LI-ion battery
- DIY frames

Technical feasibility

- **Technology:-**Flight controller, communication module, navigation and sensors
- **Expertise:-**Knowledge of mechanical, electronic and coding.
- **Infrastructure:-**CNC machines, injection molding equipment, ESCs, PDBs.
- **Constraint:-**Balancing strict weight limit, short battery life, system reliability.

“Drones overall will be more impactful than I think people recognise, in positive ways to help society”

IMPACTS



Conversational Impact

Providing rapid data collection, safe access to hazardous areas, and efficient delivery of emergency aid

Social Impact

improve community survival and safety during disasters by enabling rapid search, rescue, and supply delivery in hard-to-reach

Economical and Industrial Impact

significantly reduce disaster response costs and drive industrial growth by replacing expensive manned aircraft

Scientific Impact

Improve disaster response times by 35% to 50% and enhance data accuracy by 78%

National Impact

accelerating rescue operations, reducing responder risks, and delivering critical supplies to cut-off areas

Technical Impact

providing real-time data, mapping dangerous terrain, and improving response times by 35% to 50%

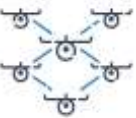


FUTURE PROSPECTS

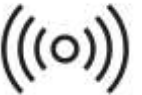
•**Autonomy:** Future drones will use artificial intelligence (AI) to fly by themselves in dangerous places without human pilots.



•**Drone Swarms:** Many drones will work together as a team. They can map large cities and search wide forests much faster.



•**Better Sensors:** New tools like LiDAR (laser mapping) and thermal cameras will help find trapped people through smoke, dark, and heavy rain.



•**Quick Deliveries:** Drones will carry life-saving medicines, blood bags, and clean water over broken roads and flooded towns.



•**Flying Cell Towers:** Drones will act as temporary signal towers to help rescue teams talk to each other when phone networks break down.



Data Sources

- <https://www.sciencedirect.com/science/article/abs/pii/S0160791X15300828>
- <https://robotics.mit.edu/design-your-own-custom-drone/>
- <https://www.intechopen.com/chapters/1154922>
- <https://iiaeit.org/drone-technology-future-avionics-engineering/>

Supporting Research Paper

- *Journal of Aviation* 9, no. 1 (2025): 225–240.
doi.org/10.30518/jav.1567696
- *Sustainable Engineering and Innovation* 6, no. 2 (2024): 199–212,
doi.org/10.37868/sei.v6i2.id309.
- in *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 50, no. 8, pp. 3059–3071, Aug. 2020, doi: 10.1109/TSMC.2018.2836922.
- Semi-autonomous drone control with safety analysis.” *Drone Systems and Applications* 11 (2023): 1–1, doi.org/10.1139/dsa-2022-0031.

Market Research

- **Global Value:** Projected to grow from \$44.32 billion in 2025 to \$53.45 billion in 2026 at a compound annual growth rate (CAGR) of 20.6%.
- **Commercial Segment:** The commercial and civil sector is growing at the fastest rate, with commercial applications expanding significantly in logistics, mapping, and agriculture.
- **Regional Leaders:** North America commands a major share (around 38% to 48%) of the global market due to heavy adoption in defense, agriculture, and infrastructure inspection.

Technical Document

- **Core Technical specification:-** Physical characteristic, performance parameters, propulsion and power, avionics & flight control.
- **Standard Documentation types:-** Aircraft maintenance manual(AMM), Illustrated Parts Catalog (IPC), Declaration of Conformity (DoC)

IMPORTANT INSTRUCTIONS



Please ensure below pointers are met while submitting the Idea PPT:

1. Kindly keep the maximum slides limit up to six **(6)**. (Including the title slide)
2. Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures
3. Keep your explanation precise and easy to understand
4. Idea should be unique and novel.
5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
6. You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.